

Midterm Examination

This is a closed calculator/computer exam. You are, however, allowed to use notes and your textbook in the exam. The last page of the exam is a Standard Normal Table, in case you need it.

You have 2 hours (120 minutes) to take the exam. The exam is 120 points, meant to roughly correspond to one point per minute of the exam. You may want to use the point allocation for each problem as an indicator for pacing yourself on the exam.

In the event of an incorrect answer, any explanation you provide of how you obtained your answer can potentially allow us to give you partial credit for a problem. For example, describe the distributions and parameter values you used, where appropriate. It is fine for your answers to include summations, products, factorials, exponentials, and combinations, unless the question specifically asks for a numeric quantity or closed form. Where numeric answers are required, the use of fractions is fine.

Problem	Score
1 (24 pts)	
2 (20 pts)	
3 (20 pts)	
4 (20 pts)	
5 (12 pts)	
6 (24 pts)	
Total (120 pts)	

THE STANFORD UNIVERSITY HONOR CODE

- A. The Honor Code is an undertaking of the students, individually and collectively:
- (1) That they will not give or receive aid in examinations; that they will not give or receive unpermitted aid in class work, in the preparation of reports, or in any other work that is to be used by the instructor as the basis of grading;
 - (2) That they will do their share and take an active part in seeing to it that others as well as themselves uphold the spirit and letter of the Honor Code.
- B. The faculty on its part manifests its confidence in the honor of its students by refraining from proctoring examinations and from taking unusual and unreasonable precautions to prevent the forms of dishonesty mentioned above. The faculty will also avoid as far as practicable, academic procedures that create temptations to violate the Honor Code.
- C. While the faculty alone has the right and obligation to set academic requirements, the students and faculty will work together to create optimal conditions for honorable academic work.

I acknowledge and accept the letter and spirit of the honor code:

Signature: _____

NAME (print): _____

1. (24 points) You are playing a card game that uses **four** standard decks of cards. There are 208 cards total.

Each deck has 52 cards (13 values each with 4 suits). Cards are only distinguishable based on their suit and value, not which deck they came from.

- (5 Points) In how many distinct ways can the cards be ordered?
- (4 points) You will be dealt the first two cards from the four decks of cards. Cards with values 10, Jack, Queen, King and Ace are considered “good” cards. What is the probability of getting two “good” cards?
- (5 points) Over the course of several rounds you observe 100 cards played. Out of the cards played only 15 were “good” cards. You are dealt the next two cards. What is the probability of getting two “good” cards now? You may assume that previously seen cards are not re-dealt.

- d. (10 Points) Consider a standard deck of 52 cards with the ace of spades on the bottom. Let the “step” be an algorithm where you take the card from the top of the deck and randomly place it in one of the 52 positions among the remaining deck including the top and bottom of the deck. If you repeat this step until the ace of spades is on the top (then execute the step one last time), the deck is perfectly shuffled. How many times do you expect to have to repeat the “step” until the deck is shuffled?

(Recall that it is fine for your answers to include summations, products, factorials, exponentials, or combinations.)

The first three parts are the basis of card counting in Black Jack. Part (d) is the starting point in Stanford professor Persis Diaconis' numerical analysis on randomness of card shuffles.

2. (20 points) You randomly sample 11 numbers from a set of 101 **distinct** numbers.
- a. (6 Points) **How many ways** can you sample 11 numbers such that the median of the sample is equal to the median of the original set? A median is the number separating the higher half of a data-set from the lower half. For example, the median of $\{0, 1, 4, 5, 9, 10, 100\}$ is 5.
- b. (4 points) What is the **probability** that the median of the sample is equal to the median of the original set?

- c. (10 Points) Let us denote the probability you calculated from Part B as p . We draw 100 independent samples (with 11 numbers each) from the original set. What is the probability that less than 10 of those samples will have the same median as the original set? Leave your answer in terms of p .

The median can be a more stable statistic than the mean

3. (20 points) The probability that a Netflix user likes a movie T_i from the “Tearjerker” genre:

Given that they **like** the Tearjerker genre is p_i .

Given that they **do not like** the Tearjerker genre is q_i .

60% of Netflix users like the Tearjerker genre.

Netflix assumes that, given a user’s preference for the genre, liking movie T_i and T_j are **conditionally independent** events. You may express all your answers in terms of q s and p s.

- a. (4 Points) What is the probability that a user likes movies T_1 , T_2 **and** T_3 given that they like the Tearjerker genre?

- b. (6 Points) What is the probability that they like movies T_1 , T_2 **or** T_3 given that they like the Tearjerker genre?

- c. (10 points) What is the probability that they like the Tearjerker genre given that they like T_1 , T_2 **and** T_3 .

Three movies in the Tearjerker genre that the CS109 team likes are: Forest Gump, The Boy in the Striped Pajamas and Good Will Hunting. We're kinda old school like that. This is the formulation behind the machine learning algorithm called Naïve Bayes. You will learn the machine learning part soon!

4. (20 points) You are part of a group at Stanford researching whether or not it would be possible to have a clean energy grid that is generated exclusively by wind.
- a. (4 Points) Let X be the amount of energy generated when the wind is blowing for a particular wind turbine. $X \sim N(\mu = 2, \sigma^2 = 64)$ mWh. Given that the wind is blowing, what is the probability that the wind farm produces more than 4 mWh of energy? Provide a numerical answer.
- b. (6 Points) The population on the grid is 40,000 people. In a given hour each person has an independent probability of 0.5 of using electricity from the grid. What is the approximate probability that more than 20,300 people use electricity in a given hour?

- c. (10 Points) The wind blows at two wind farms independently with probability p_1 , and p_2 respectively. The amount of electricity produced, **given that the wind is blowing** is distributed differently for each wind farm:

Wind farm 1: Amount of energy when the wind blows $W_1 \sim N(\mu_1, \sigma_1^2)$ gWh

Wind farm 2: Amount of energy when the wind blows $W_2 \sim N(\mu_2, \sigma_2^2)$ gWh

If the wind doesn't blow at a wind farm, that farm produces zero gWh.

What is the probability that the sum of electricity produced by the two wind farms is less than θ gWh?

Arun Majumdar's lab at Stanford is working on modernizing the electric grid. The winds of freedom blow especially hard in that lab. Wind generates a ton of electricity but it is highly stochastic.

5. (12 points) An autonomous car has two different instruments for determining its direction:

The first instrument reports a direction $D_1 = T + X_1$

The second instrument reports a direction $D_2 = T + X_2$

Where $X_1 \sim N(\mu = 0, \sigma^2 = 1)$ and $X_2 \sim N(\mu = 0, \sigma^2 = 4)$ are independent Gaussian noise and T is the true direction of the autonomous car. Before checking the instruments, the car believes that all directions in the range 50 to 60 degrees are equally likely.

What is the **probability density function** for the true direction given that the first instrument reports 57 degrees ($D_1 = 57$) and the second instrument reports 59 degrees ($D_2 = 59$)? Use a constant K in your function.

6. (24 points) You are making a mobile phone ride sharing application where people travelling the same route can share a car. A particular route is requested on average 2 times in any 5-minute period. Each request is independent.
- a. (7 Points) A user requests the route and you commit a car to take her. What is the probability that at least one other user will request the same route in the next five minutes? *Hint: it should be the same as the probability of getting at least one user without the original request.*
- b. (10 Points) A user requests the route and you commit a car to take her. All users who request the route in the next five minutes will be added to the car -- as long as the car has space. The car can fit up to three users. You will make \$6 for each user in the car (your revenue) minus \$7 (the operating cost). How much do you expect to make from this trip?

- c. (7 points) What is the probability that it takes longer than 8 minutes to get one more request to the route?

This is the idea behind Lyft Line and Uber Pool.

Midterm Solution

This handout goes over the solutions to the midterm. Overall the class performed above expectation – but still (almost) everyone walked out with room for improvement. By the final you will want to make sure you are comfortable with all the problems on the midterm.

1 Counting Cards

a

This is a permutation with indistinguishable elements question. There are 208 cards. Each value (of which there are 52) is repeated four times, because it shows up in each of four decks:

$$\text{Answer} = \frac{208!}{(4!)^{52}}$$

b

Think of this as a two step experiment. The probability of getting two good cards is the probability that the first card you get is a good card (G_1) times the probability that the second card you get is a good card (G_2):

$$\begin{aligned}\text{Answer} &= P(G_1) \cdot P(G_2) \\ &= \frac{5 \cdot 4 \cdot 4}{208} \cdot \frac{5 \cdot 4 \cdot 4 - 1}{207} \\ &= 0.147\end{aligned}$$

c

This is just like the previous question, except that the number of total cards in the deck is 100 less and the number of “good” cards in the deck is 15 less.

$$\begin{aligned}\text{Answer} &= P(G_1) \cdot P(G_2) \\ &= \frac{5 \cdot 4 \cdot 4 - 15}{108} \cdot \frac{5 \cdot 4 \cdot 4 - 16}{107} \\ &= 0.360\end{aligned}$$

d

Let X_i be a random variable which is the number of steps until the i th success. A success is the event that in a “step”, the top card gets placed below the ace of spades.

The probability of a particular success depends on i . At first, when $i = 1$, the probability of a success is $1/52$ because only one position is below the ace of spades. When $i = 2$ the probability of a success is $2/52$ since there are now two positions below the ace of spades. The probability of a success in general is $i/52$.

The number of steps until a success with known probability can be modeled as a Geometric Random Variable. Thus:

$$X_i \sim \text{Geo}(i/52).$$

Let X be the total number of steps until 52 successes, plus the single extra step to shuffle the ace of spades:

$$X = \left(\sum_{i=1}^{52} X_i \right) + 1$$

$$E[X] = E\left[\sum_{i=1}^{52} X_i\right] + 1 = \left(\sum_{i=1}^{52} E[X_i] \right) + 1$$

$$= \left(\sum_{i=1}^{52} \frac{52}{i} \right) + 1$$

The last term can be reworked into a Harmonic series. Evaluating it would be nice, but unnecessary.

2 Sampling

a

This is the combination of three steps. You must have the median in your sample. Then you need five numbers from below the median (of which there are 50 numbers) and five numbers from above the median (of which there are again 50 numbers).

$$\text{Answer} = \binom{50}{5} \cdot \binom{50}{5}$$

b

$$\text{Answer} = \frac{\binom{50}{5} \cdot \binom{50}{5}}{\binom{101}{11}}$$

c

Let X be the number of samples with the same median. $X \sim \text{Bin}(100, p)$

$$P(X < 10) = \sum_{i=0}^9 P(X = i)$$

$$= \sum_{i=0}^9 \binom{100}{i} p^i (1-p)^{100-i}$$

Given that the value of n is large and p is very small, it would be reasonable to approximate the count using a Poisson random variable $A \sim \text{Poi}(100p)$.

$$P(X < 10) \approx P(A < 10) = \sum_{i=0}^9 P(X = i) = \sum_{i=0}^9 \frac{(100p)^i}{i!} e^{-(100p)}$$

3 Netflix

a

$$\text{Answer} = p_1 \cdot p_2 \cdot p_3$$

b

Let $L_i|G$ be the event that they like movie T_i given that they like the genre.

$$\begin{aligned}\text{Answer} &= P(L_1|G \cup L_2|G \cup L_3|G) \\ &= 1 - P(L_1|G \cup L_2|G \cup L_3|G)^C \\ &= 1 - P(L_1^C|G \cdot L_2^C|G \cdot L_3^C|G) \\ &= 1 - (1 - p_1)(1 - p_2)(1 - p_3)\end{aligned}$$

Another approach is to use the inclusion/exclusion principle to expand the first line of the previous answer:

$$\begin{aligned}\text{Answer} &= P(L_1|G \cap L_2|G \cap L_3|G) \\ &= p_1 + p_2 + p_3 - (p_1 p_2) - (p_1 p_3) - (p_2 p_3) + (p_1 p_2 p_3)\end{aligned}$$

c

Solve this question using Bayes theorem:

$$\begin{aligned}P(G|L_1 L_2 L_3) &= \frac{P(L_1 L_2 L_3|G)P(G)}{P(L_1 L_2 L_3|G)P(G) + P(L_1 L_2 L_3|G^C)P(G^C)} \\ &= \frac{(p_1 p_2 p_3)(0.6)}{(p_1 p_2 p_3)(0.6) + (q_1 q_2 q_3)(0.4)}\end{aligned}$$

4 Wind Power

a

$$\begin{aligned}P(X > 4) &= 1 - P(X < 4) \\ &= 1 - F_X(4) \\ &= 1 - \Phi\left(\frac{4-2}{\sqrt{64}}\right) \\ &= 1 - \Phi\left(\frac{1}{4}\right) \\ &= 1 - 0.5987 \\ &= 0.4013\end{aligned}$$

b

Approximate the binomial with a Normal that matches the mean and variance of the binomial.

$$\begin{aligned}\mu &= np = 40,000 \cdot \frac{1}{2} = 20,000 \\ \sigma^2 &= np(1-p) = 40,000 \cdot \frac{1}{2} \cdot \frac{1}{2} = 10,000 \\ \sigma &= \sqrt{(10,000)} = 100\end{aligned}$$

When approximating normally we use the continuity correction. In this case it doesn't matter because the numbers are so big. Let Y be the number of people using electricity. $Y \sim N(\mu = 20,000, \sigma^2 = 10,000)$

$$\begin{aligned}
P(Y > 20,300) &= 1 - P(Y < 20,300) \\
&= 1 - F_Y(20,300) \\
&= 1 - \Phi\left(\frac{20,300 - 20,000}{\sqrt{10,000}}\right) \\
&= 1 - \Phi\left(\frac{300}{100}\right) \\
&= 1 - 0.9987 \\
&= 0.0013
\end{aligned}$$

c

Let Z be the amount of wind produced by the two wind farms. It is easier to incorporate the fact that wind may or may not blow at each wind-farm if you break the probability of wind being produced into mutually exclusive cases of wind in both farms, wind in farm-1, wind in farm-2 and no wind. Let E_1 be the event that wind is blowing at farm-1 and let E_2 at farm-2.

$$\begin{aligned}
P(Z < \theta) &= P(Z < \theta, E_1 E_2) + P(Z < \theta, E_1 E_2^C) + P(Z < \theta, E_1^C E_2) + P(Z < \theta, E_1^C E_2^C) \\
&= P(Z < \theta | E_1 E_2) P(E_1 E_2) \\
&\quad + P(Z < \theta | E_1 E_2^C) P(E_1 E_2^C) \\
&\quad + P(Z < \theta | E_1^C E_2) P(E_1^C E_2) \\
&\quad + P(Z < \theta | E_1^C E_2^C) P(E_1^C E_2^C) \\
&= p_1 p_2 [F_{W_1 + W_2}(\theta)] \\
&\quad + p_1 (1 - p_2) [F_{W_1}(\theta)] \\
&\quad + (1 - p_1) p_2 [F_{W_2}(\theta)] \\
&\quad + (1 - p_1) (1 - p_2) \\
&= p_1 p_2 \Phi\left(\frac{\theta - \mu_1 - \mu_2}{\sigma_1^2 + \sigma_2^2}\right) \\
&\quad + p_1 (1 - p_2) \Phi\left(\frac{\theta - \mu_1}{\sigma_1^2}\right) \\
&\quad + (1 - p_1) p_2 \Phi\left(\frac{\theta - \mu_2}{\sigma_2^2}\right) \\
&\quad + (1 - p_1) (1 - p_2)
\end{aligned}$$

Germany tried to turn their grid 100% into wind. As they learned, it is more important to chose locations where wind is generated independently then it is to chose a wind farm location where wind generates a lot of electricity.

5 Autonomous Car

This problem required a few steps. First use Bayes theorem to rewrite the probability density function. Then recognize that two of the three terms are constant. Finally plug in the density functions for the independent

instrument readings given the true direction.

$$\begin{aligned}
f(T = t | D_1 = 57, D_2 = 59) &= \frac{f(D_1 = 57, D_2 = 59 | T) f(T)}{f(D_1 = 57, D_2 = 59)} \\
&= K f(D_1 = 57, D_2 = 59 | T = t) \\
&= K f(D_1 = 57 | T = t) \cdot f(D_2 = 59 | T = t) \\
&= K \cdot \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{(57-t)^2}{2}} \right) \left(\frac{1}{2\sqrt{2\pi}} e^{-\frac{(59-t)^2}{2 \cdot 4}} \right) \\
&= K \cdot e^{-\frac{(57-t)^2}{2}} \cdot e^{-\frac{(59-t)^2}{8}}
\end{aligned}$$

There are a few ways to figure out $f(D_1 = 57 | T = t) \cdot f(D_2 = 59 | T = t)$. The one used above requires recognizing that adding a constant to a normal random variable produces a new normal with a shifted mean. The direction instruments are conditionally independent of one another given the true direction since all of their randomness comes from the independent noise terms. The other approach is to plug in the equation for the D variables and solve for the noise term X . The noise terms are defined as independent.

6 Uber Pool

a

Let X be the number of users that request the route in the next 5 minutes. $X \sim \text{Poi}(2)$.

$$\begin{aligned}
P(X \geq 1) &= 1 - P(X = 0) \\
&= 1 - e^{-2}
\end{aligned}$$

b

Let X be the number of *extra* riders beyond the first that join the car and let Z be the money that you make from the trip. Your profit function is $g(x) = 6 \cdot x - 1$:

$$\begin{aligned}
E[Z] &= P(X = 0)g(0) + P(X = 1)g(1) + P(X \geq 2)g(2) \\
&= P(X = 0)[-1] + P(X = 1)[5] + P(X \geq 2)[11] \\
&= \left(\frac{2^0}{0!} e^{-2} \right) [-1] + \left(\frac{2^1}{1!} e^{-2} \right) [5] + P(X \geq 2)[11]
\end{aligned}$$

Now we calculate $P(X \geq 2)$, because getting more than three requests will still mean that we fill the car:

$$\begin{aligned}
P(X \geq 2) &= 1 - P(X = 0) - P(X = 1) \\
&= 1 - (e^{-2}) - \left(\frac{2^1}{1!} e^{-2} \right)
\end{aligned}$$

Putting the two together:

$$\begin{aligned}
E[Z] &= \left(\frac{2^0}{0!} e^{-2} \right) [-1] + \left(\frac{2^1}{1!} e^{-2} \right) [5] + P(X \geq 2)[11] \\
&= \left(\frac{2^0}{0!} e^{-2} \right) [-1] + \left(\frac{2^1}{1!} e^{-2} \right) [5] + \left(1 - (e^{-2}) - \left(\frac{2^1}{1!} e^{-2} \right) \right) [11] \\
&= (e^{-2})[-1] + (2e^{-2})[5] + (1 - (e^{-2}) - (2e^{-2})) [11] \\
&= -e^{-2} + 10e^{-2} + 11 - 11e^{-2} - 22e^{-2} \\
&= 11 - 24e^{-2}
\end{aligned}$$

In case you are curious this works out to \$7.75

c

Let Y be the number of minutes until a user requests. You could either model Y as a geometric random variable or as an exponential.

If you model Y as an exponential, first calculate the rate per minute, which is $\frac{2}{5}$. Thus $Y \sim \text{Exp}(\lambda = \frac{2}{5})$.

$$\begin{aligned} P(Y > 8) &= 1 - P(Y < 8) \\ &= 1 - F_Y(8) \\ &= 1 - (1 - e^{-\frac{2 \cdot 8}{5}}) \\ &= e^{-\frac{16}{5}} \end{aligned}$$

If you model Y as a geometric distribution, you first have to calculate the probability that you get a request in one minute. The probability of a “success” in one minute is the probability that we worked out in part a: $1 - e^{-2/5}$. Thus $Y \sim \text{Geo}(1 - e^{-2/5})$:

$$\begin{aligned} P(Y > 8) &= 1 - P(Y < 8) \\ &= 1 - \sum_{i=1}^8 P(Y = i) \\ &= 1 - \sum_{i=1}^8 \left[1 - (1 - e^{-2/5}) \right]^{i-1} (1 - e^{-2/5}) \\ &= 1 - \sum_{i=1}^8 \left[e^{-2/5} \right]^{i-1} (1 - e^{-2/5}) \end{aligned}$$

This also works out to be $e^{-\frac{16}{5}}$. Just in case you are curious the evaluated answer is: 0.041